Statistics (Set A) — MCQ Worksheet

Mathematics • Chapter 14 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. The mean of n observations x_1, x_2, …, x■ is:

(A) (x_1 + x_2 + … + x■)/n
(B) x_1 × x_2 × … × x■
(C) Largest value
(D) Sum × n

Q2. The mean of grouped data using direct method is:

(A) Σ(f■x■)/Σf■
(B) Σx■/n
(C) Σf■
(D) Mode

Q3. In assumed mean method, the mean is:

(A) a + (Σf■d■/Σf■)
(B) a × Σf■d■
(C) a – Σf■d■
(D) Σd■/n

Q4. In step-deviation method d■ becomes:

(A) x■ – a
(B) x■ + a
(C) (x■ – a)/h
(D) (x■ + a)/h

Q5. Median is the value that lies in the middle when data is arranged in:

(A) Random order
(B) Ascending or descending order
(C) No order
(D) Reverse order

Q6. Mode is the value with the:

(A) Highest frequency
(B) Lowest frequency
(C) Middle frequency
(D) Random frequency

Q7. Empirical relation between mean, median and mode is:

(A) Mode = 3 Median – 2 Mean
(B) Mode = 2 Mean – Median
(C) Mean = Mode + Median
(D) Mode = Mean – Median

Q8. For grouped data, median = l + ((n/2 – cf)/f) × h where l is:

(A) Lower limit of median class
(B) Mean
(C) Mode
(D) Upper limit

Q9. For grouped data, mode = l + ((f_1 – f_0)/(2f_1 – f_0 – f_2)) × h where f_1 is:

(A) Frequency of preceding class
(B) Frequency of modal class
(C) Frequency of succeeding class
(D) Cumulative frequency

Q10. Mean of first 10 natural numbers is:

(A) 5
(B) 5.5
(C) 10
(D) 11

Q11. Mean of 5, 10, 15, 20, 25 is:

(A) 10
(B) 15
(C) 20
(D) 25

Q12. Median of 11, 12, 14, 13, 15 (arranged) is:

(A) 11
(B) 12
(C) 13
(D) 15

Q13. For an even number of observations, median is the:

(A) Largest
(B) Smallest
(C) Average of two middle values
(D) Mode

Q14. Mode of 2, 3, 5, 5, 7, 8 is:

(A) 2
(B) 3
(C) 5
(D) 7

Q15. In a cumulative frequency curve (ogive), the median can be found by drawing a line from cumulative frequency value of:

(A) n
(B) n/2
(C) n/4
(D) 2n

Q16. Less-than ogive plots:

(A) Upper limits vs cumulative frequency
(B) Lower limits vs cumulative frequency
(C) Midpoints vs frequency
(D) Frequency vs class

Q17. More-than ogive plots:

(A) Upper limits vs cumulative frequency
(B) Lower limits vs cumulative frequency
(C) Midpoints vs frequency
(D) Frequency vs class

Q18. The middlemost value of a data is its:

(A) Mean
(B) Median
(C) Mode
(D) Range

Q19. Class mark of class 10–20 is:

(A) 10
(B) 20
(C) 15
(D) 30

Q20. If the class size is h and lower class boundary l, the class mark equals:

(A) l + h/2
(B) l – h
(C) l × h
(D) 2l + h

Q21. Mean is most affected by:

(A) Mode
(B) Extreme values (outliers)
(C) Median
(D) Range

Q22. In a moderately skewed distribution, mean – mode ≈

(A) 3 (mean – median)
(B) 2 (mean – median)
(C) mean – median
(D) 0

Q23. Σf■ in grouped data represents:

(A) Total frequency
(B) Sum of x's
(C) Mean
(D) Mode

Q24. If the mean of 5, 7, 9, x is 8, then x =

(A) 10
(B) 11
(C) 12
(D) 9

Q25. For a symmetric distribution mean = median = mode. This is true:

(A) Always
(B) Only sometimes
(C) Never
(D) Only for skewed

— End of Worksheet —